

KUZ' MOVKA, Russian Federation

WMO: 237880

Lat: 62.3145N

Lon: 92.1132E

Elev: 203

StdP: 14.59

Time Zone: 7.00 (E07)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -55.8	(c) -50.2	(d) -60.6	(e) 0.1	(f) -55.8	(g) -54.9	(h) 0.2	(i) -50.1	(j) 13.8	(k) 7.6	(l) 11.9	(m) 9.1	(n) 0.2	(o) 90	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 24.0	(c) 84.6	(d) 65.8	(e) 80.9	(f) 64.2	(g) 76.8	(h) 63.1	(i) 68.6	(j) 79.6	(k) 66.5	(l) 76.7	(m) 64.5	(n) 73.4	(o) 4.0	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 64.8	(b) 92.5	(c) 72.4	(d) 62.9	(e) 86.4	(f) 70.5	(g) 60.9	(h) 80.5	(i) 69.0	(j) 33.0	(k) 80.1	(l) 31.3	(m) 76.8	(n) 29.8	(o) 73.1	(p) 78.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 13.1	(b) 11.1	(c) 9.3	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.5	-19.6	-9.4	10.0	26.1	39.1	55.7	63.0	55.7	43.6	25.3	0.5	-9.8
	DBStd	30.60	19.82	17.17	15.09	10.97	7.51	9.21	5.98	6.19	7.69	10.68	19.17	19.85
	HDD50	10526	2158	1664	1241	716	350	50	1	20	222	764	1486	1854
	HDD65	15207	2623	2084	1706	1166	804	297	108	292	643	1229	1936	2319
	CDD50	864	0	0	0	0	11	222	403	198	30	0	0	0
	CDD65	71	0	0	0	0	0	19	46	6	0	0	0	0
	CDH74	1369	0	0	0	0	5	515	735	109	5	0	0	0
	CDH80	350	0	0	0	0	0	135	199	16	0	0	0	0
Wind	WSAvg	3.0	1.9	2.1	2.8	4.3	4.0	3.5	3.0	3.0	3.2	3.8	2.7	2.1
Precipitation	PrecAvg	29.5	2.0	1.6	1.7	1.9	2.2	2.6	2.6	3.1	2.8	3.5	3.0	2.5
	PrecMax	37.7	3.6	2.6	3.0	3.0	4.1	5.0	5.2	5.0	4.8	5.6	6.0	4.2
	PrecMin	20.4	0.2	0.7	0.6	0.7	0.9	0.6	0.5	1.0	0.4	0.9	1.5	1.1
	PrecStd	4.5	0.8	0.5	0.6	0.7	0.8	0.9	1.0	1.0	1.0	0.8	1.0	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.4	29.3	42.1	54.7	72.4	87.8	89.3	82.1	71.8	46.7	34.6	29.2
		MCWB	23.1	28.1	34.5	42.2	54.4	66.2	68.7	65.8	60.3	41.4	32.8	27.8
	2%	DB	19.8	25.7	36.6	49.7	64.2	84.0	85.7	76.4	65.1	43.7	30.9	25.2
		MCWB	18.7	24.1	32.3	39.0	50.6	63.9	67.5	63.8	56.4	39.3	29.4	23.9
	5%	DB	14.7	20.4	33.5	44.6	57.4	80.6	82.2	73.1	59.1	40.3	28.6	22.1
		MCWB	13.5	19.0	30.5	37.0	47.0	62.6	65.8	62.3	53.4	37.9	27.2	21.0
	10%	DB	10.5	15.4	31.0	40.9	52.3	75.7	78.5	69.0	55.5	37.1	26.1	18.2
		MCWB	9.4	14.3	28.4	34.5	43.2	61.2	63.5	60.3	50.9	35.3	25.0	17.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.4	28.6	35.8	43.0	56.8	69.6	72.1	68.8	61.7	42.8	33.1	28.1
		MCDB	24.3	29.2	39.7	52.8	69.3	80.8	83.6	76.8	70.3	45.2	34.1	29.2
	2%	WB	18.8	24.5	32.8	40.0	52.0	67.1	69.9	65.9	57.5	39.9	29.8	24.2
		MCDB	19.8	25.9	35.7	47.1	61.6	78.8	82.1	73.8	62.5	42.6	31.0	25.1
	5%	WB	13.5	19.1	30.8	37.7	48.3	64.7	67.7	63.8	54.4	37.7	27.2	21.0
		MCDB	14.6	20.2	33.4	43.1	56.7	75.8	78.3	70.6	58.8	40.1	28.4	22.0
	10%	WB	9.5	14.2	28.6	35.7	44.0	62.2	65.7	61.5	51.2	35.5	25.0	17.1
		MCDB	10.6	15.2	30.7	40.1	51.2	73.2	75.5	67.8	54.8	37.2	26.2	18.1
Mean Daily Temperature Range	5% DB	MDBR	17.0	19.6	24.7	21.7	19.4	25.3	24.0	21.3	16.5	11.7	17.0	17.3
		MCDBR	16.7	16.6	19.9	25.9	30.2	35.1	32.5	29.7	23.9	13.0	15.7	12.8
		MCWBR	16.2	16.2	16.5	17.0	17.4	16.4	16.0	17.9	15.1	10.2	15.1	12.1
	5% WB	MCDBR	16.3	16.1	17.4	22.5	27.9	28.6	27.4	25.4	22.1	12.5	14.9	12.8
		MCWBR	15.9	15.8	14.8	15.5	17.2	14.4	14.9	16.4	14.6	10.1	14.5	12.3
Clear-Sky Solar Irradiance	taub		0.256	0.297	0.331	0.356	0.372	0.387	0.432	0.411	0.351	0.307	0.241	0.229
	taud		1.901	1.970	1.982	2.016	2.163	2.382	2.246	2.268	2.398	2.193	1.951	1.865
	Ebn at Noon		113	194	236	258	266	265	247	241	237	195	119	63
	Edh at Noon		13	25	36	43	41	34	38	34	25	20	11	7
All-Sky Solar Radiation	RadAvg		88	351	781	1287	1611	1749	1568	1077	643	262	115	41
	RadStd		13	31	82	147	147	182	147	114	95	50	21	6

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	+5.33	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page